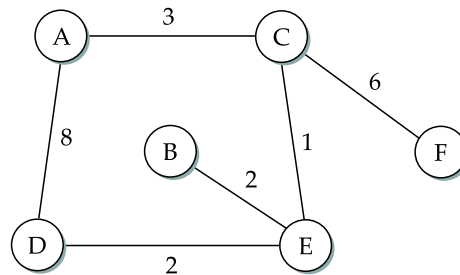


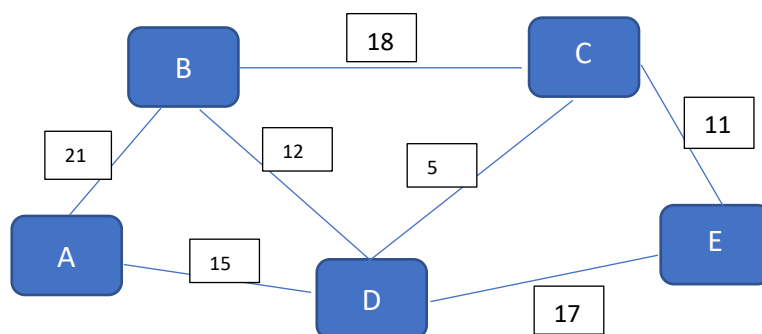
IT694
PRACTISE PROBLEMS

1. For the network given in Figure 4.50, show how the link-state algorithm builds the routing table for node D.



STEP	N	D(A),p(A)	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	D	8	INF	INF	0	2	INF
1	[D,E]	6	4	3			
2	[D,E,C]	6	4				9
3	[D,E,C,B]	6					9
4	[D,E,C,B,A]						9
5	[D,E,C,B,A,F]						

2. Let us say DAIICT IT help desk bought new servers. They decided to run a distance-vector protocol for routing between these servers (even though it is a rather small network). The link costs are given.



The initial routing table at node A is:

<i>Destination</i>	<i>Cost</i>	<i>Next Hop</i>
B	21	B
C	∞	---
D	15	D
E	∞	---

a. Show the initial routing table at E

<i>Destination</i>	<i>Cost</i>	<i>Next Hop</i>
A	INF	-
B	INF	-
C	11	C
D	17	D

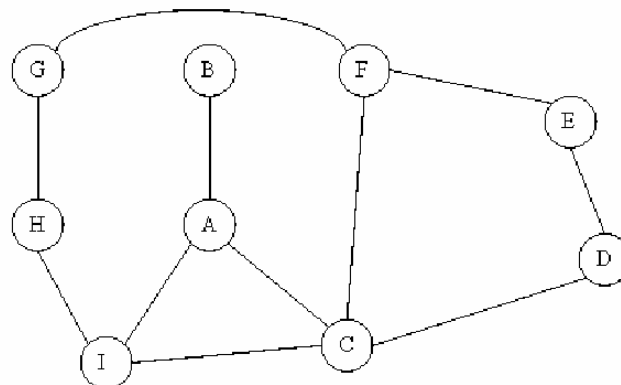
b. Show the routing table of node E after one iteration of the algorithm:

<i>Destination</i>	<i>Cost</i>	<i>Next Hop</i>
A	$17 + 15 = 32$	D
B	$11 + 18 = 29$ OR $17 + 12 = 29$	C / D
C	11	C
D	16	C

c. Show the routing table of node E after two iterations of the algorithm

<i>Destination</i>	<i>Cost</i>	<i>Next Hop</i>
A	$16 + 15 = 31$	C (via D)
B	$16 + 12 = 28$	C
C	11	C
D	16	C

3. Consider the network configuration shown in figure below. Assume that each link has the same cost.



a) Run Bellman-Ford algorithm on this network to compute the routing table for node A. Show A's distance to all other nodes at each step.

Solution:

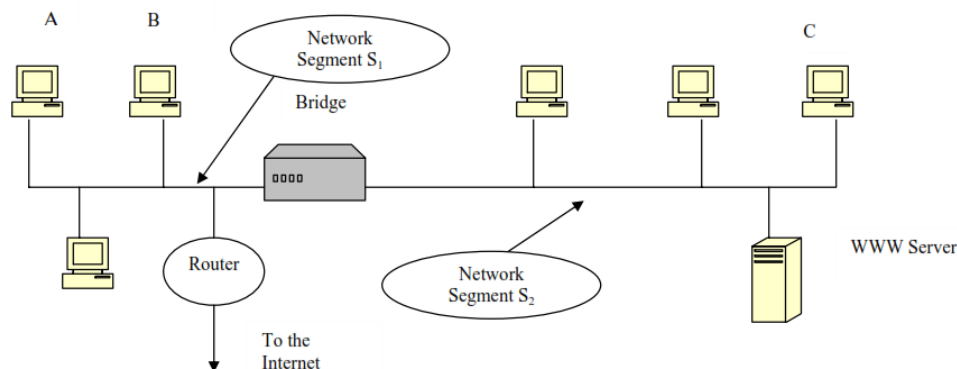
A	B	C	D	E	F	G	H	I
0	1	1	∞	∞	∞	∞	∞	1
0	1	1	2	∞	2	∞	2	1
0	1	1	2	3	2	3	2	1

b) Suppose the link A-B goes down. As a result, A advertises a distance of infinity to B. Describe in detail a scenario where C takes a long time to learn that B is unreachable

Solution:

- A advertises a distance of infinity to B, but, C and I advertise a distance of 2 to B.
- Depending on the ordering (i.e. timing) of these messages, this might happen: I upon knowing that B can be reached from C with distance 2, concludes that it can reach B with a distance of 3 (via C) and advertises this to A.
- A concludes that it can reach B with a distance of 4 (via I), and advertises this to C.
- C concludes that it can reach B with a distance of 5 (via A).
- This continues forever if the distances are unbounded. This is called the count-to-infinity problem.

Consider the following local area network below and answer the questions



- Do the hosts A and C share the same network part of their IP addresses?
ANS: Yes. Host A can directly access host C at the link layer, as the learning switch is transparent to all the hosts.
- Can host C access the WWW server at the same time as host B accesses the Internet through the router? Explain.
ANS: Yes. The switch will not forward the traffic from network segment S₁ in this case, and the two network segments are partitioned into two independent collision domains.

- c. Can host A access the WWW server at the same time as host B accesses the Internet through the router? Explain.

ANS: No. Host A needs to use the same network segment (S) upon which Host B and the WWW server is communicating on.

- d. What is the maximum size of the switch's table (in the number of entries) in this configuration?

ANS: 8

- e. When host A sends an ARP query to learn the data link address of the router, can host C hear it? Explain.

ANS: Yes. Host A sends a BROADCAST frame out and the switch will forward all the broadcast frames that it receives to the other segment.

IP Subnetting already shared.